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CRISP-DM: Step-1 | Business Understanding

OBJECTIVE:

This activity is designed to explore the initial phases of the CRISP-DM framework – Business Understanding,

Data, Understanding, and Data Preparation – through scenario-based analysis. It involves identifying real-

world problems in the banking sector, examining a publicly available dataset, and preparing the data for future

modeling. The exercise emphasizes the importance of aligning data mining efforts with clearly defined

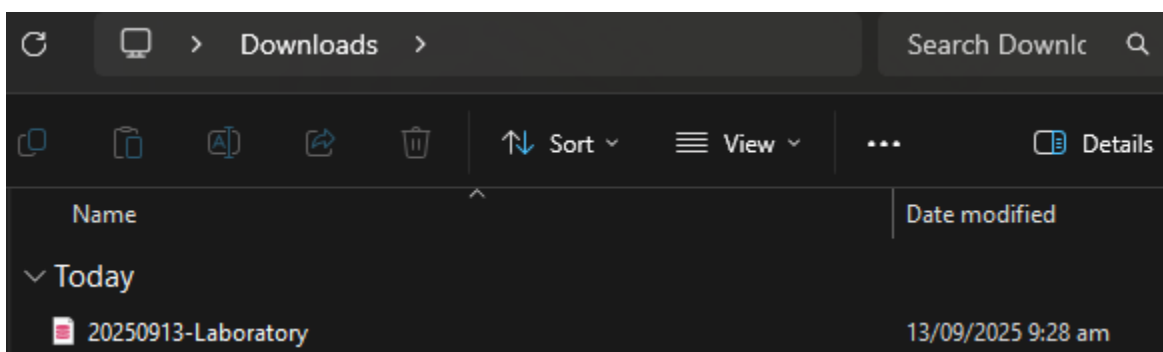
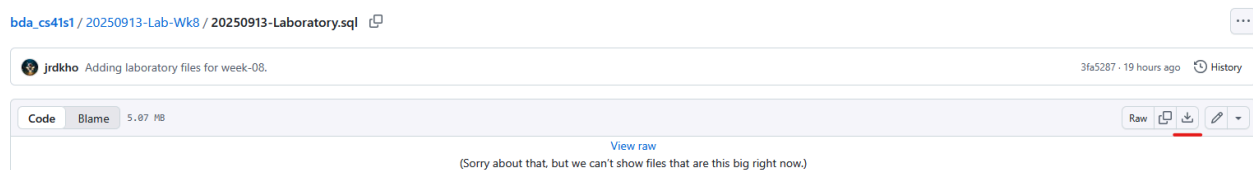
business goals, resource considerations, and contextual constraints, forming a foundation for informed

decision-making and effective analytical planning.

CRISP-DM: Step-2 | Data Understanding

INSTRUCTIONS:

1.) Download the MySQL data-dump located at https://github.com/AI-Artisans/bda_cs41s1/tree/main/20250913-Lab-Wk8

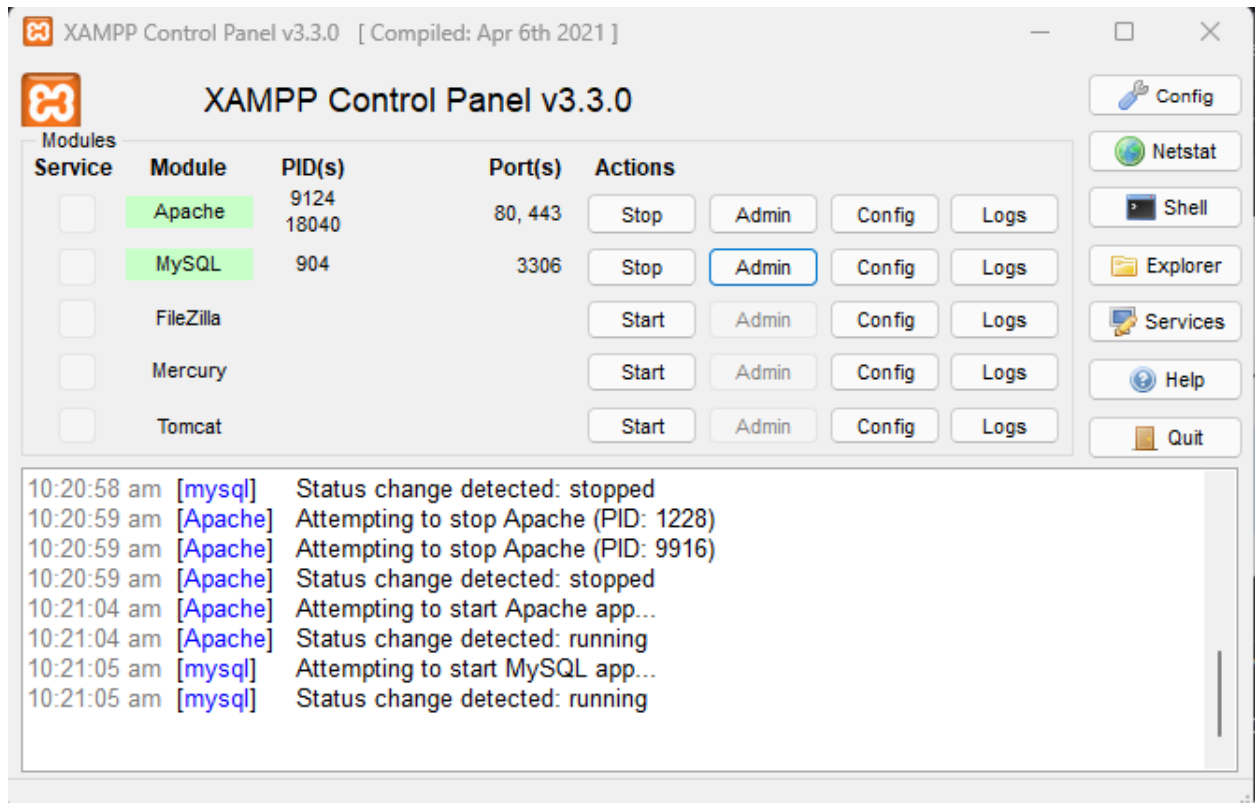


2.) From the command line (cmd.exe), restore the MySQL data-dump using the command below:

`mysql -u [MySQL DB User] [MySQL database] < [local path of MySQL data-dump file]`

example:

`mysql -u root testdb < D:\20250913-Laboratory.sql`



3.) After the data-dump restoration, explore the newly created table and its contents.

Server: 127.0.0.1 » Database: testdb » Table: bank_marketing																		
Browse Structure SQL Search Insert Export Import Privileges Operations Tracking Triggers																		
← → id age job marital education default_status balance housing loan contact day month duration campaign pdays previous outcome target																		
☐				1	58	management	married	tertiary	no	2143.00	yes	no	unknown	5 may	261	1	-1	0 unknown no
☐				2	44	technician	single	secondary	no	29.00	yes	no	unknown	5 may	151	1	-1	0 unknown no
☐				3	33	entrepreneur	married	secondary	no	2.00	yes	yes	unknown	5 may	76	1	-1	0 unknown no
☐				4	47	blue-collar	married	unknown	no	1506.00	yes	no	unknown	5 may	92	1	-1	0 unknown no
☐				5	33	unknown	single	unknown	no	1.00	no	no	unknown	5 may	198	1	-1	0 unknown no
☐				6	35	management	married	tertiary	no	231.00	yes	no	unknown	5 may	139	1	-1	0 unknown no
☐				7	28	management	single	tertiary	no	447.00	yes	yes	unknown	5 may	217	1	-1	0 unknown no
☐				8	42	entrepreneur	divorced	tertiary	yes	2.00	yes	no	unknown	5 may	380	1	-1	0 unknown no
☐				9	58	retired	married	primary	no	121.00	yes	no	unknown	5 may	50	1	-1	0 unknown no
☐				10	43	technician	single	secondary	no	593.00	yes	no	unknown	5 may	55	1	-1	0 unknown no
☐				11	41	admin.	divorced	secondary	no	270.00	yes	no	unknown	5 may	222	1	-1	0 unknown no
☐				12	29	admin.	single	secondary	no	390.00	yes	no	unknown	5 may	137	1	-1	0 unknown no
☐				13	53	technician	married	secondary	no	6.00	yes	no	unknown	5 may	517	1	-1	0 unknown no
☐				14	58	technician	married	unknown	no	71.00	yes	no	unknown	5 may	71	1	-1	0 unknown no
☐				15	57	services	married	secondary	no	162.00	yes	no	unknown	5 may	174	1	-1	0 unknown no
☐				16	51	retired	married	primary	no	229.00	yes	no	unknown	5 may	353	1	-1	0 unknown no
☐				17	45	admin.	single	unknown	no	13.00	yes	no	unknown	5 may	98	1	-1	0 unknown no
☐				18	57	blue-collar	married	primary	no	52.00	yes	no	unknown	5 may	38	1	-1	0 unknown no
☐				19	60	retired	married	primary	no	60.00	yes	no	unknown	5 may	219	1	-1	0 unknown no
☐				20	33	services	married	secondary	no	0.00	yes	no	unknown	5 may	54	1	-1	0 unknown no
☐				21	28	blue-collar	married	secondary	no	723.00	yes	yes	unknown	5 may	262	1	-1	0 unknown no
☐				22	56	management	married	tertiary	no	779.00	yes	no	unknown	5 may	164	1	-1	0 unknown no
☐				23	32	blue-collar	single	primary	no	23.00	yes	yes	unknown	5 may	160	1	-1	0 unknown no
☐				24	25	services	married	secondary	no	50.00	yes	no	unknown	5 may	342	1	-1	0 unknown no
☐				25	40	retired	married	primary	no	0.00	yes	yes	unknown	5 may	181	1	-1	0 unknown no

REFLECTION:

- What challenges did you experience during the MySQL data-dump restoration? (5 pts)
 - A lot of challenges I encountered such as cmd.exe prompt errors because I don't know the administrator's password and can't get through the steps. So I used XAMPP as an alternative and there I have the choice to use MySQL in the XAMPP.
- With respect to the newly created MySQL table, complete the details below (5 pts):

Number of Attributes : 18 Columns/Attributes

Number of Objects: 45211 Rows/Objects

- With respect to the objects inside the table, complete the details below (10 pts):

Attribute	Data Type
ID	Int(11)
Age	Int(11)
Job	Varchar(50)
Marital	Varchar(20)

Education	Varchar(50)
Default_Status	Varchar(10)
Balance	Decimal(10,2)
Housing	Varchar(5)
Loan	Varchar(5)
Contact	Varchar(20)
Day	Tinyint(3) Unsinged
Month	Varchar(10)
Duration	Int(11)
Campaign	Int(11)
PDays	Int(11)
Previous	Int(11)
Outcome	Varchar(20)
Target	Varchar(10)

CRISP-DM: Step-3 | Data Preparation

INSTRUCTIONS:

1.) Perform initial data exploration using the restored data-dump. Complete item #1 in Reflection section

Server: 127.0.0.1 » Database: testdb » Table: bank

Browse Structure SQL Search

Run SQL query/queries on table testdb.bank_marketing: ?

```

1  -- Job summary
2  SELECT job, COUNT(*) AS freq
3  FROM bank_marketing
4  GROUP BY job
5  ORDER BY freq DESC;
6
7  -- Marital summary
8  SELECT marital, COUNT(*) AS freq
9  FROM bank_marketing
10 GROUP BY marital
11 ORDER BY freq DESC;
12
13 -- Education summary
14 SELECT education, COUNT(*) AS freq
15 FROM bank_marketing
16 GROUP BY education
17 ORDER BY freq DESC;
18
19 -- Contact summary
20 SELECT contact, COUNT(*) AS freq
21 FROM bank_marketing
22 GROUP BY contact
23 ORDER BY freq DESC;
24

```

job	freq ▾ 1
blue-collar	9732
management	9458
technician	7597
admin.	4996
services	3639
retired	2170
self-employed	1579
entrepreneur	1487
unemployed	1303
housemaid	1240
student	938
svcs	515
unknown	277
Adm	175
Ret	94
???	11

marital	freq ▾ 1
married	27214
single	12790
divorced	4624
Div	583

contact	freq ▼ 1
cellular	28947
unknown	13020
telephone	2719
cell.	338
Tel	187

education	freq ▼ 1
secondary	23082
tertiary	13272
primary	6851
unknown	1843
2nd	120
3rd	29
???	14

2.) Based on the summary produced in Reflection item #1, kindly perform data clean-up on the following

attributes:

a. Job

✓ 269 rows affected. (Query took 0.0003 seconds.)

```
UPDATE bank_marketing SET job = LOWER(job);
```

[[Edit inline](#)] [[Edit](#)] [[Create PHP code](#)]

✓ 277 rows affected. (Query took 0.0003 seconds.)

```
UPDATE bank_marketing SET job = NULL WHERE job IN ('unknown', '');
```

[[Edit inline](#)] [[Edit](#)] [[Create PHP code](#)]

✓ 4996 rows affected. (Query took 0.0007 seconds.)

```
UPDATE bank_marketing SET job = 'admin' WHERE job IN ('admin.', 'administrator');
```

[[Edit inline](#)] [[Edit](#)] [[Create PHP code](#)]

```

1  -- Standardize 'adm' → 'admin'
2  UPDATE bank_marketing SET job = 'admin' WHERE job = 'adm';
3
4  -- Standardize 'ret' → 'retired'
5  UPDATE bank_marketing SET job = 'retired' WHERE job = 'ret';
6
7  -- Standardize 'svcs' → 'services'
8  UPDATE bank_marketing SET job = 'services' WHERE job = 'svcs';
9
10 -- Replace ??? and NULL with 'unknown'
11 UPDATE bank_marketing SET job = 'unknown' WHERE job IS NULL OR job = '???';
12

```

b. Marital

✓ 583 rows affected. (Query took 0.0003 seconds.)

```
UPDATE bank_marketing SET marital = LOWER(marital);
```

[[Edit inline](#)] [[Edit](#)] [[Create PHP code](#)]

✓ 0 rows affected. (Query took 0.0002 seconds.)

```
UPDATE bank_marketing SET marital = NULL WHERE marital IN ('unknown','');
```

[[Edit inline](#)] [[Edit](#)] [[Create PHP code](#)]

✓ 0 rows affected. (Query took 0.0003 seconds.)

```
UPDATE bank_marketing SET marital = 'single' WHERE marital = 'singl';
```

[[Edit inline](#)] [[Edit](#)] [[Create PHP code](#)]

c. Education

✓ 0 rows affected. (Query took 0.0002 seconds.)

```
UPDATE bank_marketing SET education = LOWER(education);
```

[[Edit inline](#)] [[Edit](#)] [[Create PHP code](#)]

✓ 1843 rows affected. (Query took 0.0003 seconds.)

```
UPDATE bank_marketing SET education = NULL WHERE education IN ('unknown','');
```

[[Edit inline](#)] [[Edit](#)] [[Create PHP code](#)]

✓ 0 rows affected. (Query took 0.0002 seconds.)

```
UPDATE bank_marketing SET education = 'basic' WHERE education IN ('basic.9y','basic.6y','basic.4y');
```

[[Edit inline](#)] [[Edit](#)] [[Create PHP code](#)]

```
1 -- Replace short forms with consistent values
2 UPDATE bank_marketing SET education = 'secondary' WHERE education = '2nd';
3 UPDATE bank_marketing SET education = 'tertiary' WHERE education = '3rd';
4
5 -- Replace ??? and NULL with 'unknown'
6 UPDATE bank_marketing SET education = 'unknown' WHERE education IS NULL OR education = '???';
7
```

d. Contact

✓ 187 rows affected. (Query took 0.0003 seconds.)

```
UPDATE bank_marketing SET contact = LOWER(contact);
```

[[Edit inline](#)] [[Edit](#)] [[Create PHP code](#)]

✓ 13020 rows affected. (Query took 0.0003 seconds.)

```
UPDATE bank_marketing SET contact = NULL WHERE contact IN ('unknown','');
```

[[Edit inline](#)] [[Edit](#)] [[Create PHP code](#)]

✓ 0 rows affected. (Query took 0.0003 seconds.)

```
UPDATE bank_marketing SET contact = 'telephone' WHERE contact = 'phone';
```

[[Edit inline](#)] [[Edit](#)] [[Create PHP code](#)]


```
-- Standardize 'cell.' → 'cellular'
UPDATE bank_marketing SET contact = 'cellular' WHERE contact = 'cell.';

-- Standardize 'tel' → 'telephone'
UPDATE bank_marketing SET contact = 'telephone' WHERE contact = 'tel';

-- Replace NULL with 'unknown'
UPDATE bank_marketing SET contact = 'unknown' WHERE contact IS NULL;
```

3.) Perform another set of data exploration after the data clean-up and complete item #2 in the Reflection section.

REFLECTION:

1.) Provide the initial summary for the following attributes (10 pts):

a. Job

job	COUNT(*)
NULL	277
???	11
adm	175
admin	4996
blue-collar	9732
entrepreneur	1487
housemaid	1240
management	9458
ret	94
retired	2170
self-employed	1579
services	3639
student	938
svcs	515
technician	7597
unemployed	1303

b. Marital

marital	COUNT(*)
div	583
divorced	4624
married	27214
single	12790

c. Education

education	COUNT(*)
NULL	1843
2nd	120
3rd	29
???	14
primary	6851
secondary	23082
tertiary	13272

d. Contact

contact	COUNT(*)
NULL	13020
cell.	338
cellular	28947
tel	187
telephone	2719

2.) Provide the final summary for the following attributes (15 pts) :

a. Job

job	COUNT(*)
admin	5171
blue-collar	9732
entrepreneur	1487
housemaid	1240
management	9458
retired	2264
self-employed	1579
services	4154
student	938
technician	7597
unemployed	1303
unknown	288

b. Marital

marital	COUNT(*)
div	583
divorced	4624
married	27214
single	12790

c. Education

education	COUNT(*)
primary	6851
secondary	23202
tertiary	13301
unknown	1857

d. Contact

contact	COUNT(*)
cellular	29285
telephone	2906
unknown	13020

3.) Based on the results produced in items #1 and #2, what issues did you encounter during the clean-up process? (5 pts)

- I've encountered several clean-up processes such as, dealing with inconsistent data, some of the attributes are not quite spelled right and some of them are abbreviations. In order to fix this I have to standardize it into one uniform value. Second, ambiguous data in "job" and "education" value there is "???" objects that are unidentified and I did classified them as "unkown", and such NULLS were found that has to be identified as "unknown" as well just to maintain the integrity of data. In "education" attribute, there is "2nd" and "3rd" object that obscure our data so it has to be standardized as well. Lastly, since it is a huge volume of data I've carefully integrated SQL queries accurately and consistent across the entire table.

4.) What are the possible repercussions if data clean-up is neglected during the "Data Preparation" phase of the CRISP-DM framework? (10 pts)

- When data cleaning is not done properly at the Data Preparation stage of the CRISP-DM framework, the quality and reliability of the entire analysis may be negated. Values that are not consistently applied such as treating adm and admin, ret and retired, etc. as two different categories can result in inaccurate summaries and skewed findings. This would also be poor in predictive models, since they would be trained with noisy and inconsistent data, and it would be hard to find meaningful patterns. Further, unclear entries such as "???" and high levels of NULLs diminish the quality of data that could result in business leaders making faulty decisions, based on incomplete or unreliable data. Failure to cleanup also adds to the burden at the later stages of work since the analysts would take up more time cleaning up mistakes in the modeling or reporting rather than paying attention to insights. Finally, data clean-up is

important to ensure the accuracy, trustworthiness, and usefulness of the data mining process, and neglecting it may easily lead to severe consequences on decision-making.